

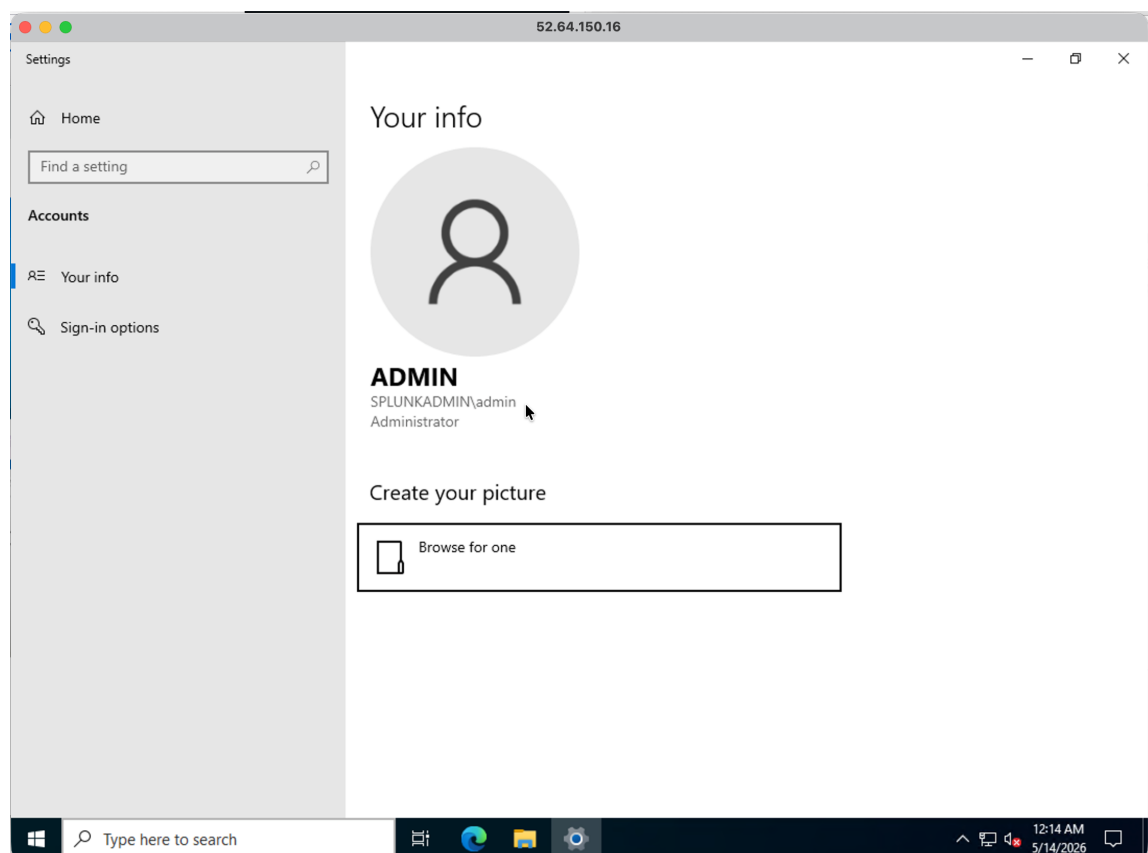
Simulating Privilege Escalation Attack on Windows Server and Client

Summary

This document contains steps related to the Simulation of Privilege Escalation attack on both Windows Server (DC) (Event ID 4728) and Domain-joined Windows Client (Event ID 4732). It explains two different scenarios – Domain Escalation and Local Escalation that can be simulated.

Scenario 1 – Simulate Domain Escalation

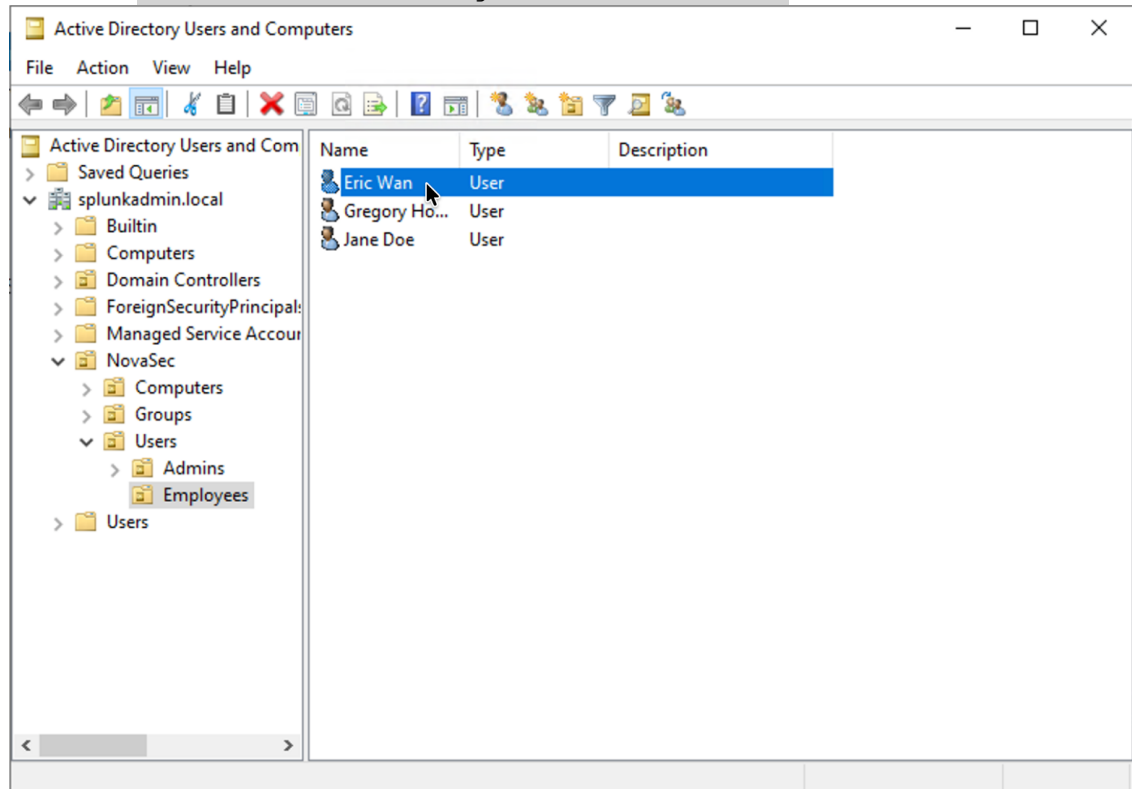
1. Login as Domain Admin into Domain Controller (Server)



2. Add a domain user to Domain Admins group

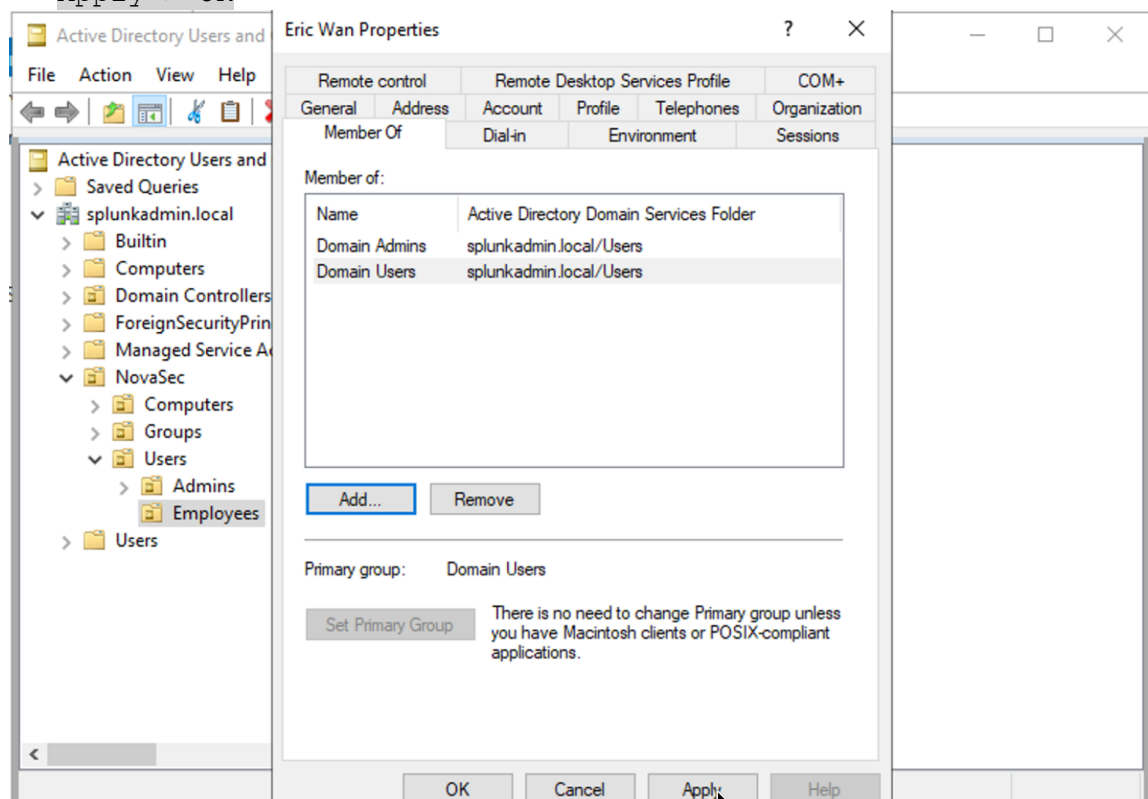
1) Open Active Directory Users and Computers (ADUC)

Go to `Start > Server Manager > Tools > ADUC`



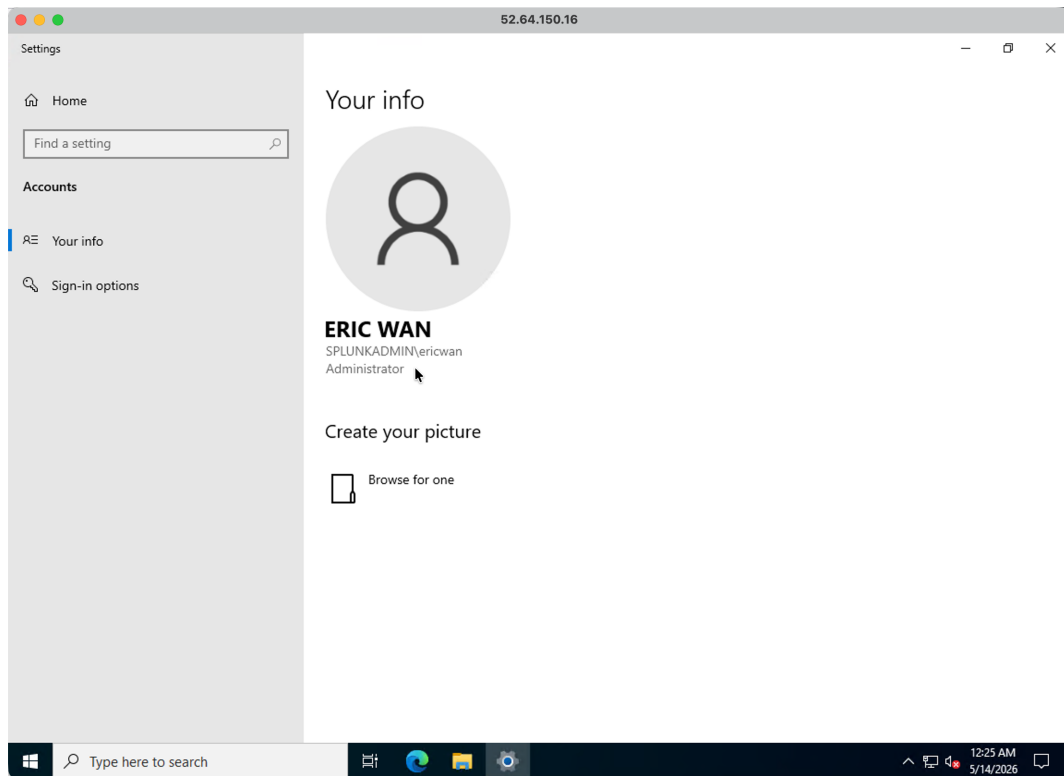
2) Add a Domain User to Domain Admins

Right-click a user in Employees OU > Properties > Member Of tab > Add... > Search "domain admins" > Check Names > OK > Apply > OK



3) Logout from Domain Admin

3. Login as the domain user with Domain Admins privileges

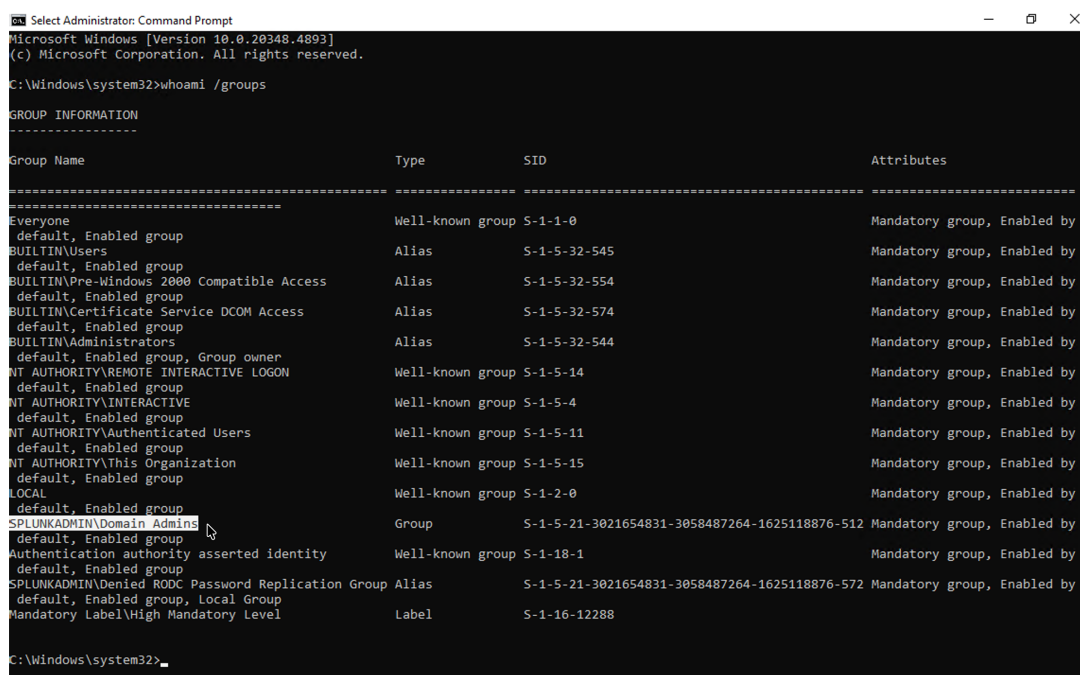


4. Test Privileges

Open Command Prompt as Administrator

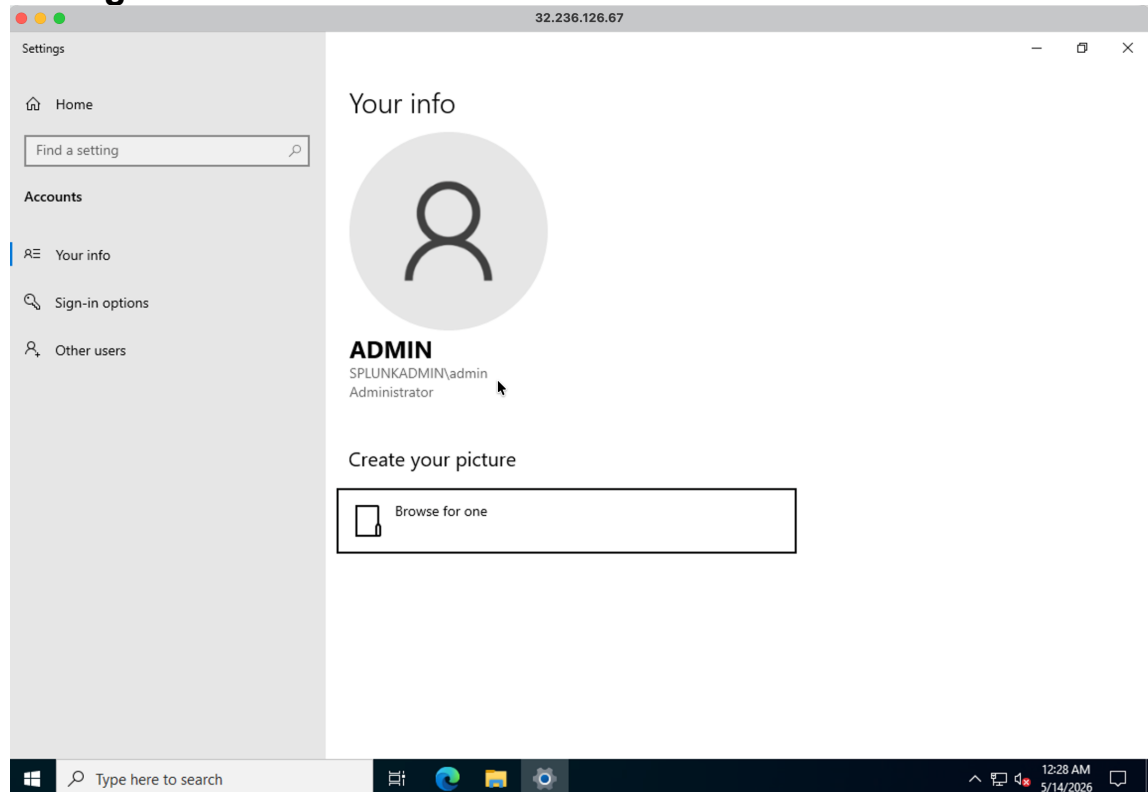
Run:

```
whoami /groups
```



Scenario 2 – Simulate Local Escalation

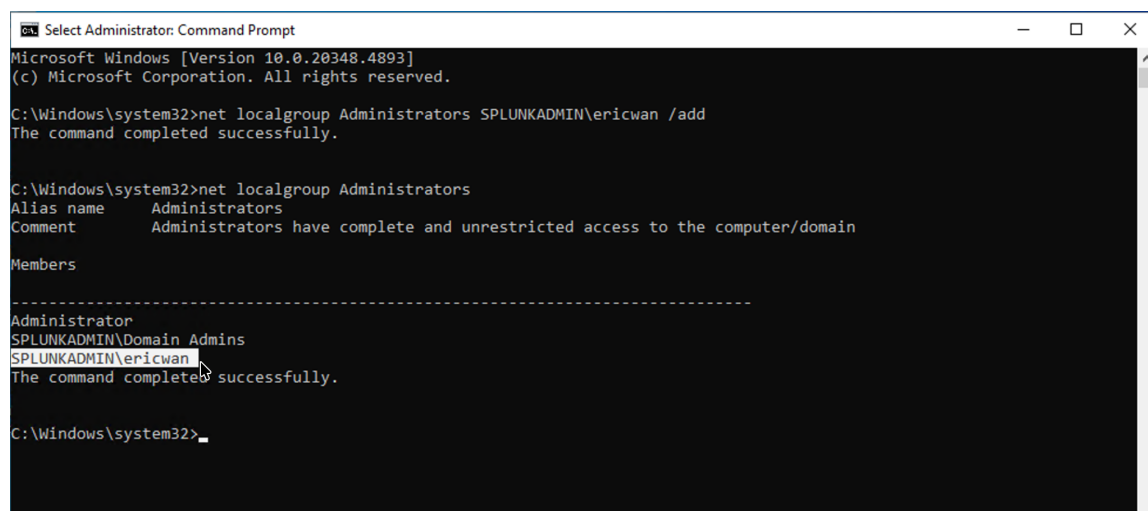
1. Login on Client as Domain Admin



2. Add the same domain user (Scenario 1) to local group Administrators

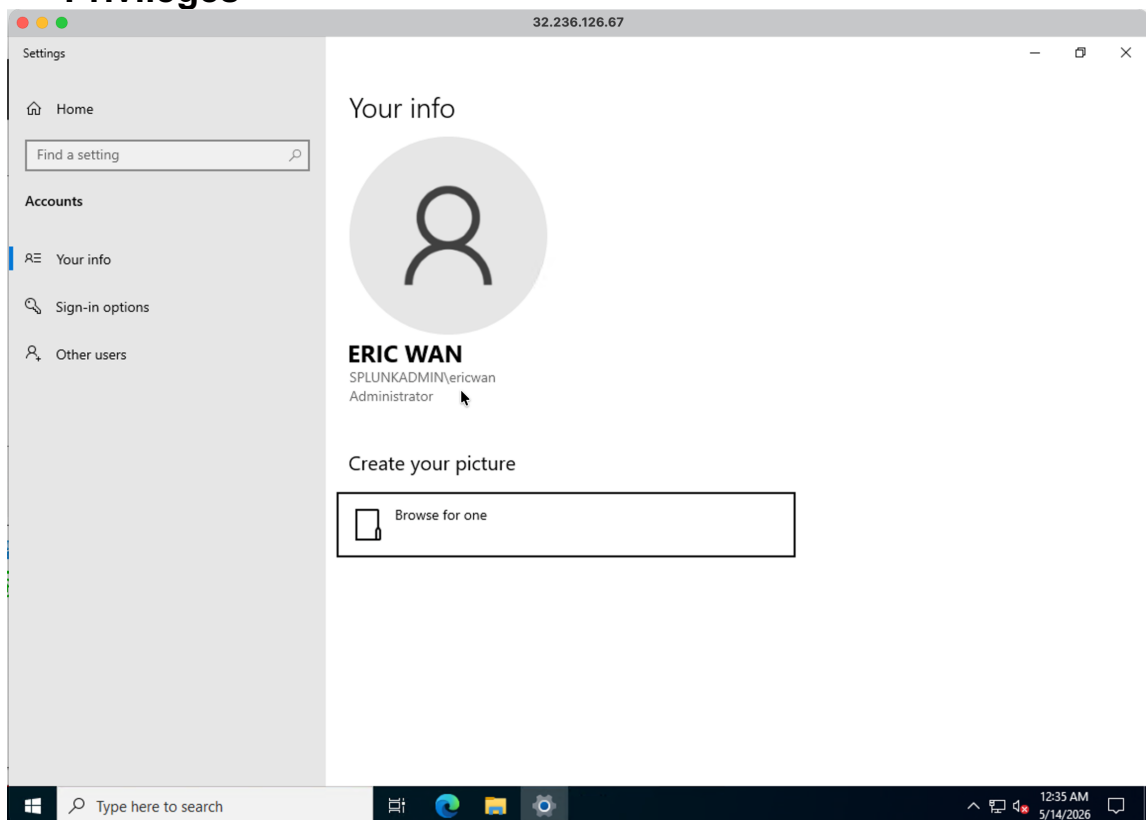
Open **Command Prompt as Administrator**,
Run:

```
net localgroup Administrators SPLUNKADMIN\<username> /add
```



3. Logout from Domain Admin

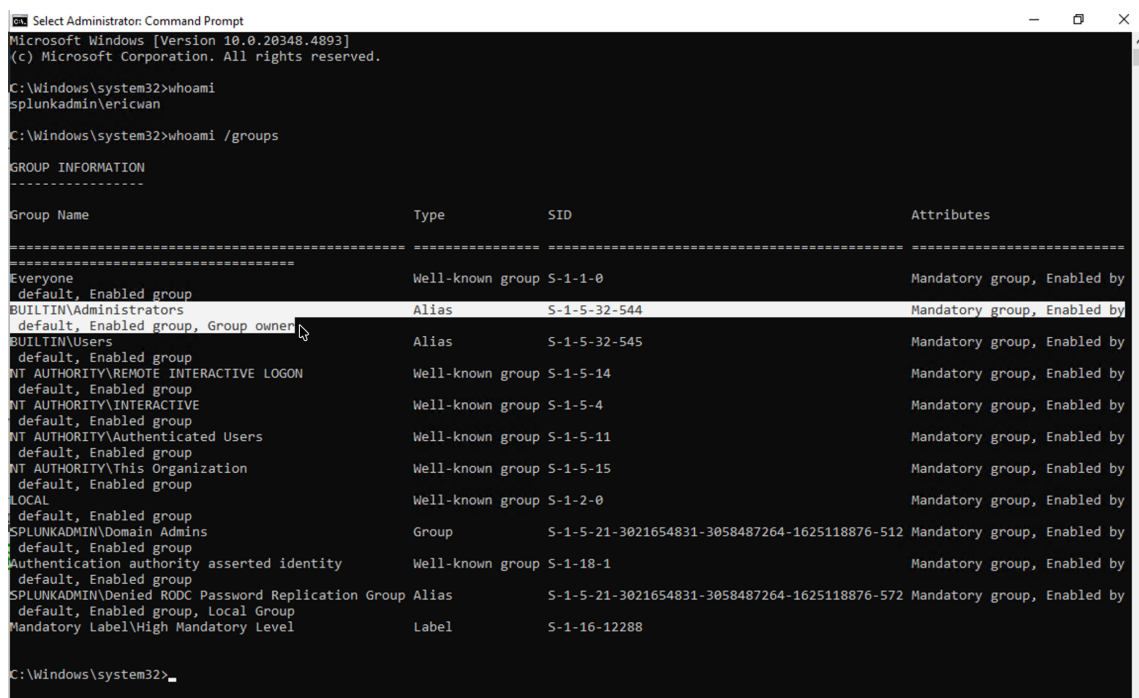
4. Login as the domain user with Local Administrators Privileges



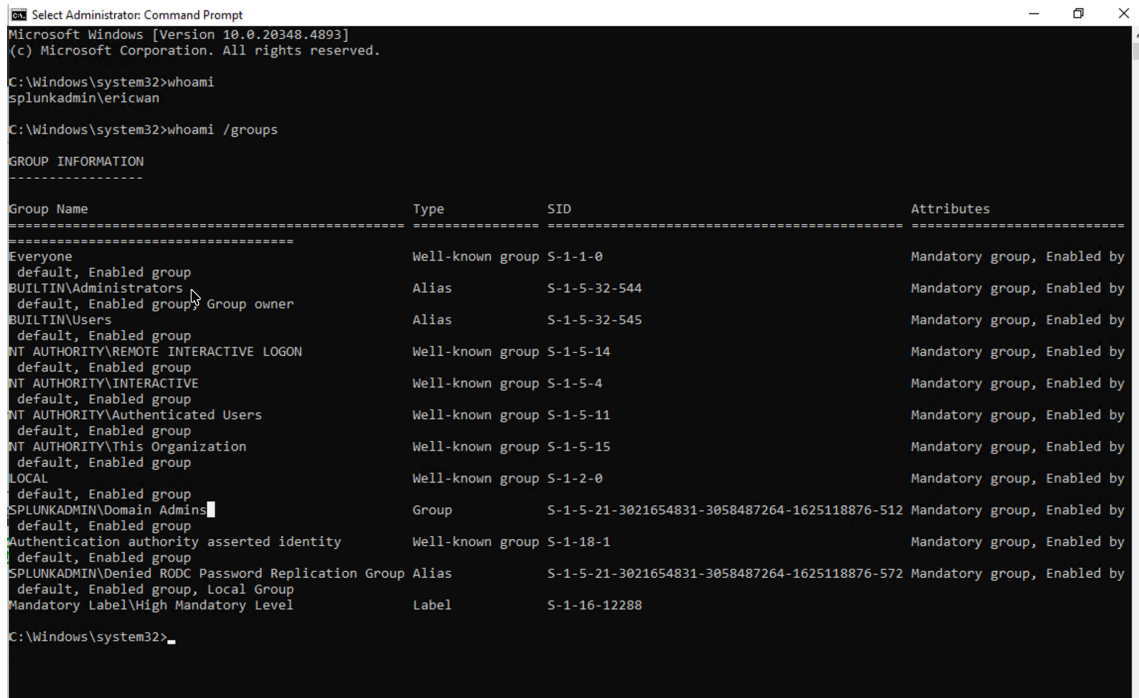
5. Test Privileges

Open **Command Prompt as Administrator**,
Run:

```
whoami /groups
```



Note: If you have **executed both the Scenarios**, then you should see **both Local Administrators and Domain Admins** here.



```

Select Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.4893]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
splunkadmin\ericwan

C:\Windows\system32>whoami /groups

GROUP INFORMATION
-----
Group Name                                     Type                SID                                     Attributes
-----
Everyone                                     Well-known group    S-1-1-0                               Mandatory group, Enabled by
default, Enabled group
BUILTIN\Administrators                       Alias               S-1-5-32-544                         Mandatory group, Enabled by
default, Enabled group, Group owner
BUILTIN\Users                               Alias               S-1-5-32-545                         Mandatory group, Enabled by
default, Enabled group
NT AUTHORITY\REMOTE INTERACTIVE LOGON       Well-known group    S-1-5-14                             Mandatory group, Enabled by
default, Enabled group
NT AUTHORITY\INTERACTIVE                     Well-known group    S-1-5-4                               Mandatory group, Enabled by
default, Enabled group
NT AUTHORITY\Authenticated Users             Well-known group    S-1-5-11                             Mandatory group, Enabled by
default, Enabled group
NT AUTHORITY\This Organization               Well-known group    S-1-5-15                             Mandatory group, Enabled by
default, Enabled group
LOCAL                                        Well-known group    S-1-2-0                               Mandatory group, Enabled by
default, Enabled group
SPLUNKADMIN\Domain Admins                    Group               S-1-5-21-3021654831-3058487264-1625118876-512 Mandatory group, Enabled by
default, Enabled group
Authentication authority asserted identity  Well-known group    S-1-18-1                             Mandatory group, Enabled by
default, Enabled group
SPLUNKADMIN\Denied RODC Password Replication Group Alias               S-1-5-21-3021654831-3058487264-1625118876-572 Mandatory group, Enabled by
default, Enabled group, Local Group
Mandatory Label\High Mandatory Level        Label               S-1-16-12288
C:\Windows\system32>

```

Next Steps

Validate the logs received from Windows DC and Domain-joined Client in Splunk Enterprise using Extraction and Detection – Refer

24_Logs_Validation_using_Extraction_Detection_for_ServerClient_Events_V1.pdf